Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 25-03-2026

Responsive E-Commerce Interface with State-Persistent Cart

SDG: SDG No – SDG9- Industry, Innovation and Infrastructure

Presented By:

MAMILLA AASHA NAGA SAI SHREE

– VTU24557 – 23UECS1128.

Courses Handling Faculty:

Dr . HAFIZUR RAHMAN

INTRODUCTION

1
2
3
4
5
6
7
8
9

- Modern online shopping demands seamless and interactive user experiences.
- Users expect their cart items to persist across sessions for convenience.
- State-persistent carts prevent loss of selected products, improving satisfaction.
- This project maintains cart state dynamically using web technologies.
- Enhances usability, reduces abandoned carts, and boosts sales.
- Real-world relevance includes platforms like Amazon, Flipkart, and Shopify.
- Integrates a responsive frontend with a backend database to store cart data.
- Allows users to add, remove, and view products with real-time updates.

PROBLEM STATEMENT

1

2

- Users often lose their cart items when leaving e-commerce sites.
- Current systems fail to maintain cart state across sessions.
- This causes inconvenience and increases abandoned purchases.
- Manually tracking selected products is inefficient and frustrating.
- A system is needed to dynamically preserve cart data in real time.
- Users should be able to add, remove, and view products seamlessly.
- The system must work across devices and different screen sizes.
- A state-persistent cart improves user experience and reduces lost sales.

OBJECTIVE(S)

1
2
3
4
5
6
7
8
9

- Design a responsive e-commerce interface.
- Implement a state-persistent shopping cart.
- Allow users to add and remove products easily.
- Preserve cart data across sessions and devices.
- Improve user experience and reduce abandoned carts.

PROPOSED SYSTEM

1

2

3

4

5

6

7

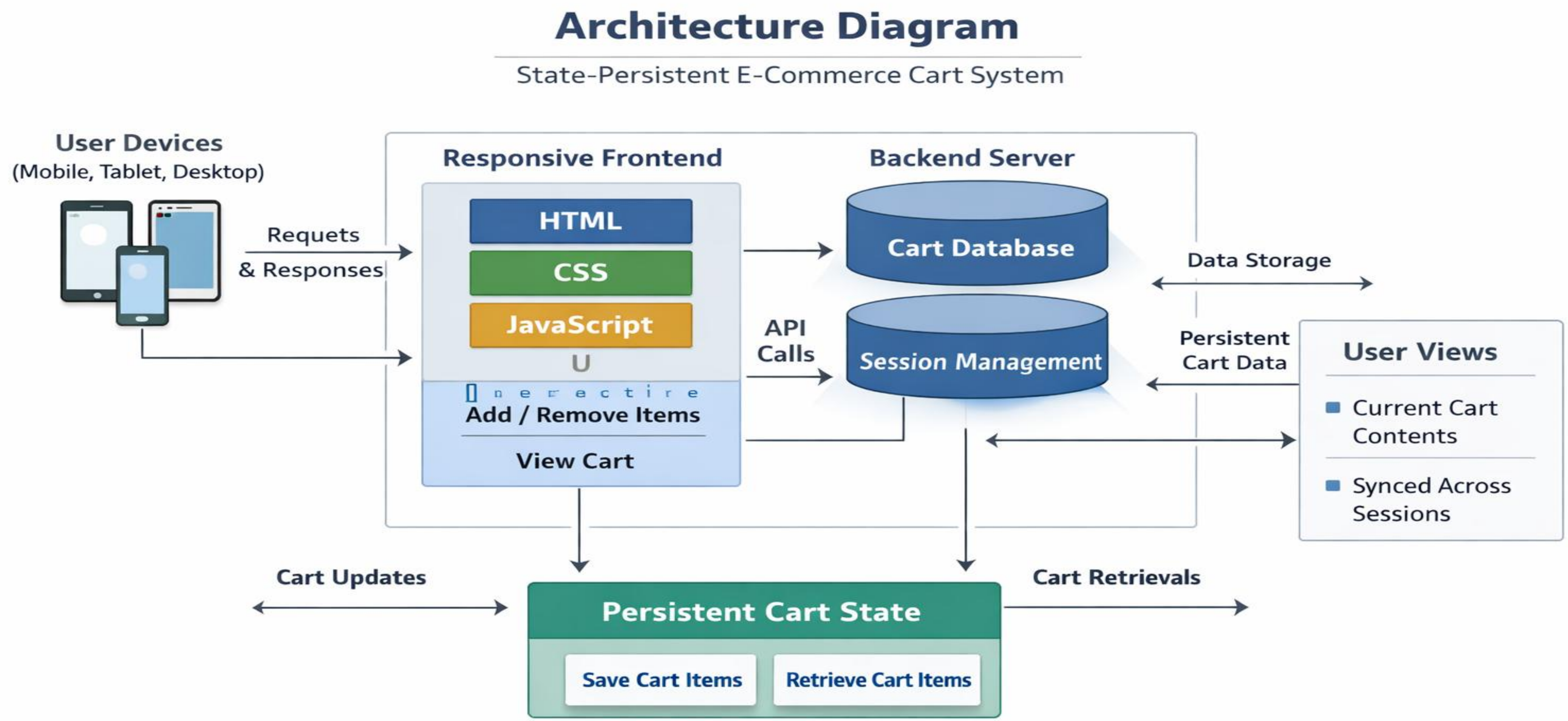
8

9

- Develop a responsive e-commerce platform with a dynamic cart system.
- Use frontend technologies (HTML, CSS, JavaScript) for interactive UI.
- Store cart data in a backend database to retain items across sessions.
- Enable users to add, remove, and view products in real time.
- Automatically save cart state to prevent loss of selected items.
- Works seamlessly across devices and screen sizes.
- Reduces abandoned carts and improves customer satisfaction.
- Provides a reliable, efficient, and user-friendly shopping experience.

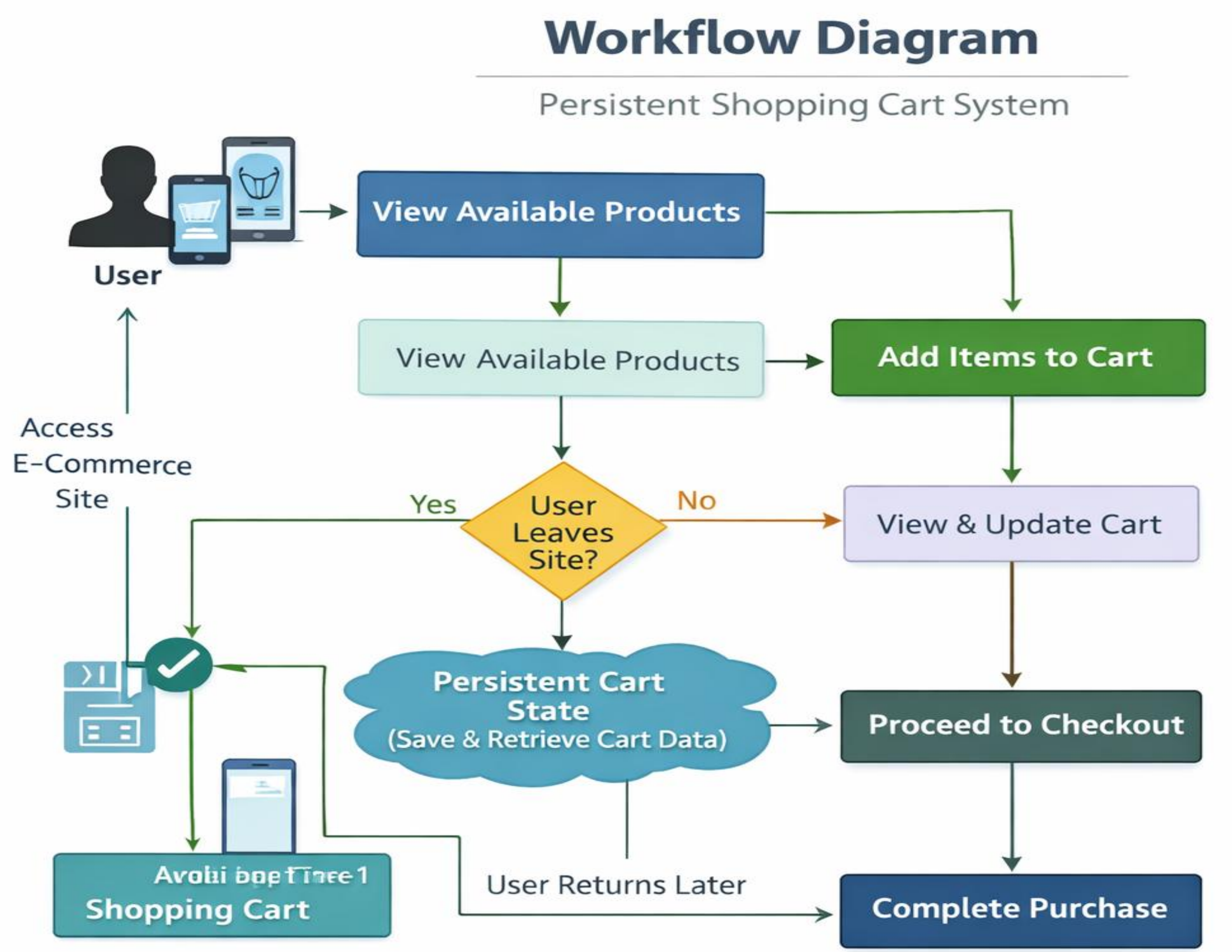
PROPOSED SYSTEM ARCHITECTURE

1
2
3
4
5
6
7
8
9



WORKFLOW

1
2
3
4
5
6
7
8
9



TECHNOLOGIES USED

- **Frontend:**
  - HTML
  - CSS
  - JavaScript
- **Backend:**
  - Node.js
- **Database:**
  - MySQL
- **Tools:**
  - Vs Code
  - XAMPP
  - **Hardware/Software Requirements:**
  - Intel i3 or above
  - 4GB RAM(Mininum)
  - Windows 10/11
  - Web Browser(Chrome/Edge)
  - XAMPP Server

1 2 3 4 5 6 7 8 9

MODULE DESCRIPTION

1
2
3
4
5
6
7
8
9

Module 1: User Management
Handles user registration, login, and profile management, including role-based access for admin and customer users.

Module 2: Product Management
Allows adding, updating, and deleting products, managing categories, stock levels, and product details.

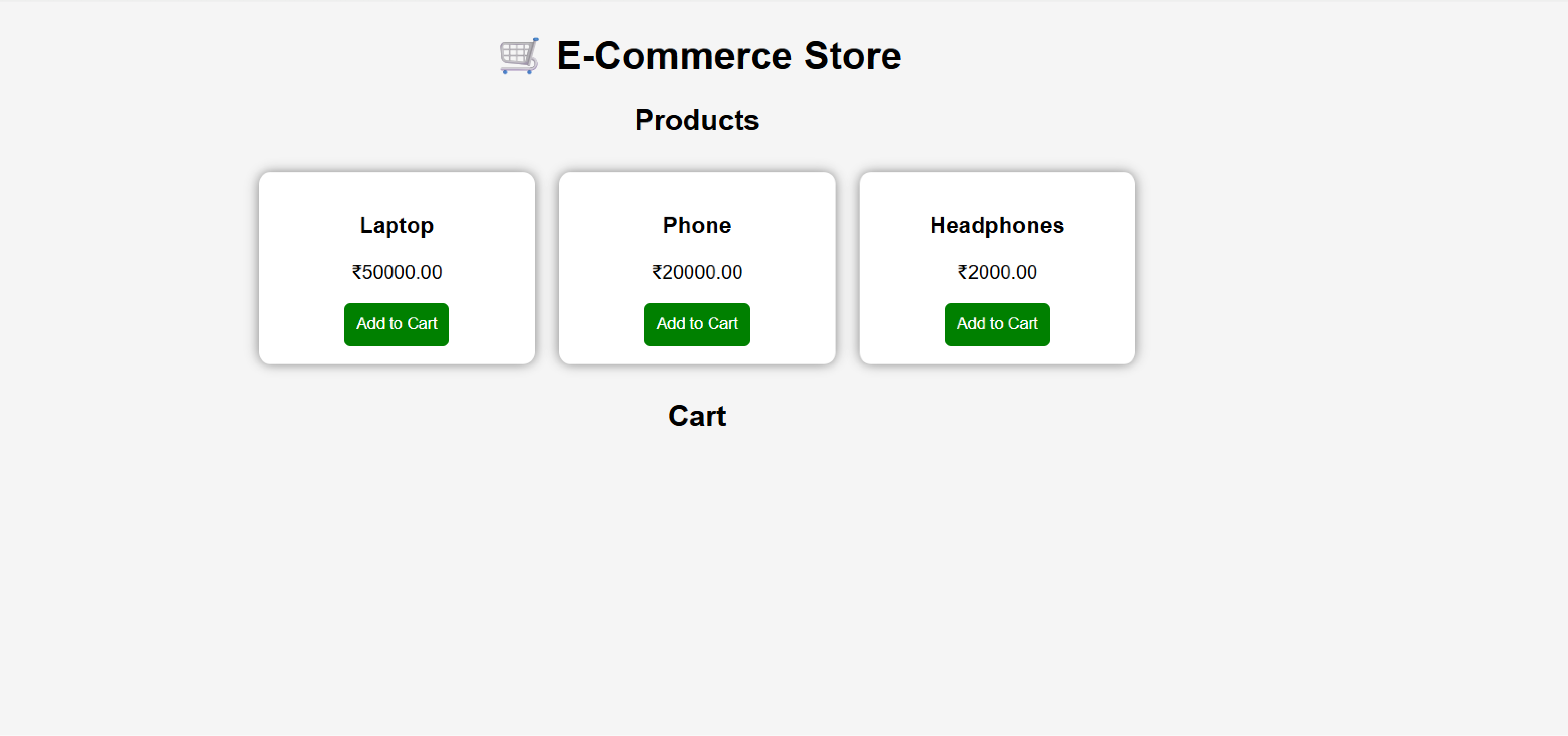
Module 3: Cart & Order Management
Manages shopping cart functionality, order placement, tracking, and checkout processes with payment integration.

Module 4: Database Management
Handles data storage, retrieval, and ensures integrity for users, products, and orders.

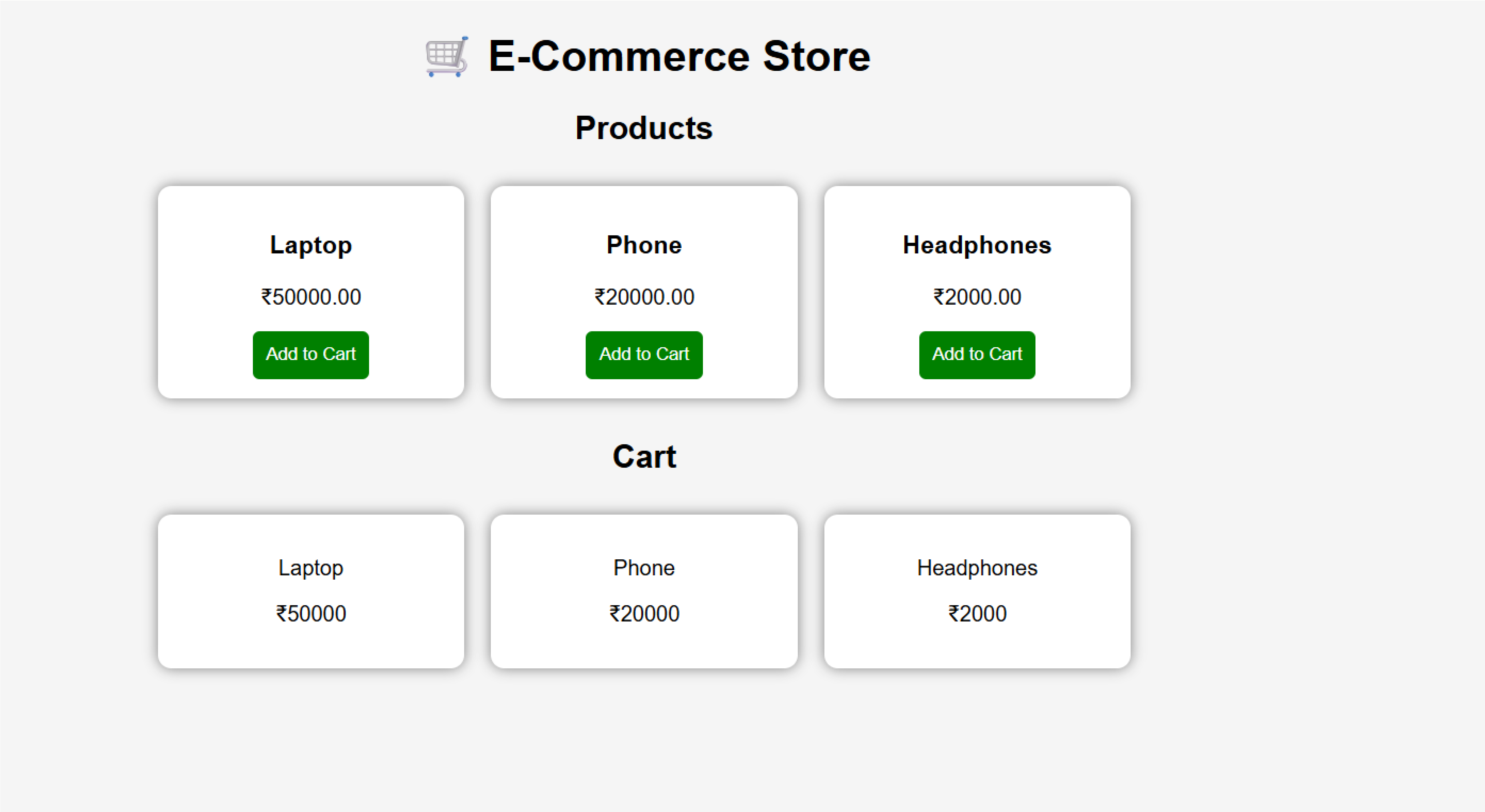
Module 5: Reporting & Analytics
Generates sales reports, user activity logs, and visual insights for better decision-making.

IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8
9

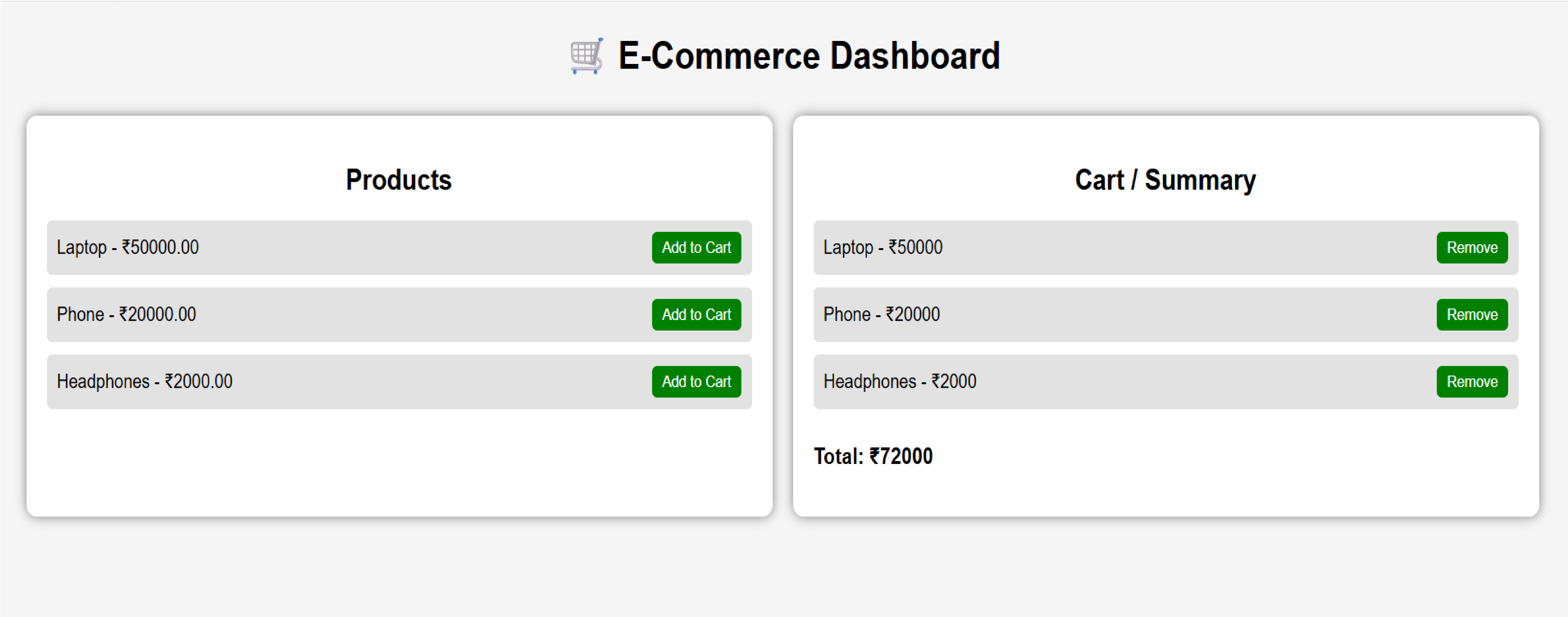


IMPLEMENTATION SCREENSHOTS



IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8(B)
9



ADVANTAGES & LIMITATIONS

Advantages:

- **Centralized Management** – Admin can manage users, products, orders, and analytics from a single dashboard.
- **Improved User Experience** – Easy navigation, shopping cart, and seamless checkout enhance customer satisfaction.
- **Real-Time Data Access** – Instant access to product availability, orders, and user activity.
- **Data-Driven Decisions** – Analytics module helps in understanding trends and making informed business decisions.
- **Secure Transactions** – User authentication and order processing ensure secure handling of data and payments.

Limitations:

- **Initial Setup Complexity** – Requires proper database design and server configuration.
- **Maintenance Required** – Regular updates needed for product catalog, security patches, and performance optimization.
- **Dependent on Internet** – Users cannot access dashboard features offline.
- **Scalability Concerns** – Large numbers of users/orders may require advanced optimization and server resources.
- **Payment Gateway Dependence** – Online transactions rely on third-party payment services which may fail occasionally.

Thank you